08. Security Baseline and Its Importance in SOC

1. What Is a Security Baseline?

Definition: A security baseline is a documented description of normal or approved system, user and network behaviour used for comparison during security monitoring.

A baseline may describe expected applications, login times, network connections, running services and administrative activity.

2. Examples of a Baseline

Area	Normal Baseline Example
User activity	Employees normally sign in during their working hours
Network traffic	Workstations connect to approved business services
Processes	Known and authorized applications run on endpoints
Services	Only approved Windows services are installed
Accounts	Administrator membership is limited to authorized personnel
DNS	Devices resolve expected business and software-update domains
Network ports	Only approved services listen on designated ports

These are examples, not universal rules. Every organization must establish its own baseline.

3. Why Is a Baseline Important?

- Helps identify deviations from normal behaviour.
- Reduces unnecessary alerts when normal activity is understood.
- Helps detect unusual login patterns and network connections.
- Supports incident investigation and threat hunting.
- Helps measure changes in system configuration.
- Provides evidence for security monitoring and auditing.

4. Example of Baseline Analysis

Scenario: A workstation normally connects to a small number of approved servers during office hours. One evening, it begins contacting many unfamiliar external IP addresses.

An analyst should:

1. Compare the activity with the normal network baseline.
2. Identify the source process and destination addresses.
3. Review DNS, firewall and endpoint logs.
4. Determine whether the activity was approved.
5. Document the evidence and escalate if necessary.

The unusual behaviour is a clue for investigation, not proof of compromise.

5. Baseline vs Anomaly

Baseline	Anomaly
Describes expected behaviour	Behaviour that differs from the baseline
Built from approved configurations and observations	Identified by comparing observed activity with expectations
Helps analysts understand normal activity	Helps analysts decide what needs investigation

6. How a SOC Analyst Uses a Baseline

A SOC analyst can establish baselines by:

- Reviewing historical network and endpoint logs.
- Identifying authorized users, services and applications.
- Documenting normal login and traffic patterns.
- Updating the baseline when approved business changes occur.
- Investigating deviations and correlating multiple sources of evidence.

7. Conclusion

A security baseline helps a SOC analyst understand what normal behaviour looks like. Comparing current activity with this baseline helps identify unusual events and prioritize investigations more effectively.